



Concept Generation and Screening Report

BM2210: Biomedical Device Design
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1 Introduction

Team VitaSync identified, in Assignment 1, the need for a way to continuously monitor the onset of anhidrosis (sweat cessation) in individuals exposed to high environmental heat or prolonged physical exertion, in order to provide an actionable early warning of heat stroke before systemic physiological collapse occurs. Heat stroke is a severe, thermoregulation related medical emergency that disproportionately affects outdoor laborers, military personnel, athletes and elderly individuals in tropical climates. Existing approaches to detecting it, such as environmental WBGT monitors, commercial smartwatches, ingestible core-temperature pills and buddy system visual observation, either fail to capture the individual's physiological state directly or are too invasive, expensive, or subjective to be deployed at scale. This gap motivates the development of a low cost, non invasive wearable capable of fusing galvanic skin response (sweat conductivity), environmental heat index, Photoplethysmography(PPG) and skin temperature data to flag the earliest signs of heat stroke risk.

This report documents the concept generation and screening process undertaken by Team VitaSync to translate this identified need into a viable device concept, following the procedure outlined in Section II of the course textbook(Biodesign). The process begins with initial concept selection, where a range of candidate solutions were generated through structured brainstorming and concept mapping. These candidate concepts were then screened against intellectual property, regulatory and business model considerations to narrow the field to the most promising options. The surviving concepts were subsequently evaluated through concept testing, involving prototyping and a structured testing plan, before a final concept was selected using a weighted selection matrix informed by an assessment of killer risks. The remainder of this report presents each stage of this process in detail, with supporting evidence for ideation and screening activities compiled in the appendices.

2 Initial Concept Selection

2.1 Brainstorming

The ideation phase followed a disciplined, iterative process focused purely on generating a high volume of concepts, deferring judgment to encourage an unrestricted flow of ideas. Our team utilized a multidisciplinary approach, combining internal brainstorming with insights from medical professionals and athletes, to address the complex stakeholder landscape.

Through iterative brainstorming sessions, we identified these sensors to use in our device.

- Humidity sensors to detect the presence of sweat.
- Ag-AgCl electrodes to measure Galvanic Skin Response (GSR) and track spikes in skin electrical resistance associated with sweat gland failure.
- Thermistors or high-accuracy medical grade infrared skin temperature sensors (e.g., MLX90614, MAX30208) to monitor core heating toward 40°C.
- Environmental sensors (BME 280, DHT 31) to capture ambient heat index and humidity.

Furthermore, our team generated the idea to incorporate a Photoplethysmography (PPG) sensor, following a consultation with a doctor.The doctor suggested this to track severe tachycardia, providing an additional physiological marker since the heart pumps faster to cool the body prior to a heatstroke.

To identify most suitable anatomical placement, we brainstormed different wearable configurations, including a full vest and a targeted back patch.First we listed potential sensor attachment sites across the body, such as the forehead, armpit, ear canal, and the upper thoracic region.Based on feedback from sportsmen, we identified specific ergonomic constraints, noting the device must have high wearability, remain thin, feature good ventilation, and have as few moving or separate parts as possible to prevent discomfort during physical exertion.

2.2 Concept Maps

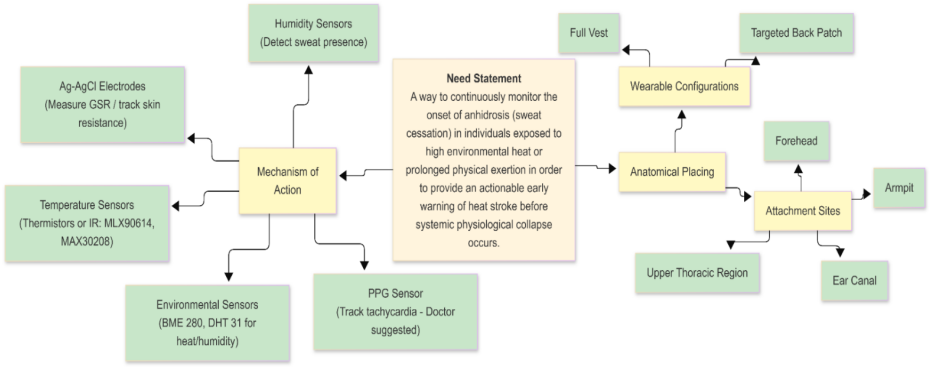


Figure 1: Concept map for the heat stroke early warning device

2.3 Screening Against Need Criteria

Concepts were evaluated considering their alignment with need statement, on their ability to continuously and accurately monitor physical exertion in high environmental heat and wearability.

We identified that using standalone sensors for specific tasks is not practical because they can give inaccurate readings based on different conditions.

- Standalone Humidity Sensors: If worn outside clothing, sweat is absorbed by the fabric, preventing accurate evaporation measurement.
- Standalone GSR: During heavy exercise, sweat completely floods the electrodes. Furthermore, as sweat evaporates, left-behind salt crystals artificially maintain high electrical conductivity even after the user stops sweating (anhidrosis), leading to critical false-negative readings.

So we moved to a multi-sensor arrangement, which gives a combined accurate final decision about heat strokes. Also we screened different anatomical locations to place sensors, considering accuracy of readings and wearability.

- Forehead - Intense exercise causes heavy sweat to flood the sensors, which ruins the skin resistance data, Placement is uncomfortable for end users like sportsmen.
- Ear canal - It is incredibly difficult to comfortably fit a multi sensor array into the ear without annoying the user, final device should be extremely small, making it hard to implement the device with a multi- sensor arrangement with the desired accuracy.

Because of the above reasons, we rejected the options forehead, armpit, and ear canal from possible anatomical locations in initial concept selection stage.

2.4 Generated Concepts Summary

Table 1: Summary of Initial Concepts Generated

No.	Concept Name	Brief Description
1	Standalone Sensor Monitoring	Using individual sensors (humidity or GSR) in isolation to detect sweat or skin resistance; rejected due to inaccuracy from sweat flooding and fabric absorption.
2	Multi-Sensor Arrangement	A system combining thermistors, ambient sensors, GSR, and PPG to provide a unified, accurate physiological assessment of heatstroke risk.
3	Head-Mounted Placements	Attaching sensors to the forehead or ear canal; rejected due to severe sweat flooding, size constraints, and user discomfort during physical exertion.
4	Torso-Mounted Configurations	Integrating the sensor array into a full vest, targeted back patch, or upper thoracic region to ensure high wearability, thinness, and ventilation.

3 Concept Screening

3.1 Intellectual Property (IP) Considerations

3.1.1 Design patent vs Utility patent

The Hardware of the Device itself is difficult to patent as it includes third party components such as ESP32 and other over the shelf components and sensors. What can be patented is the actual process of detecting Heatstroke risk, which cross references GSR circuit AC phase shift, MAX30105 data and MF52AT NTC thermistor and SHT31 temperature sensor for temperature readings. This process can be used as IP to create a Utility patent.

The physical design of the Product including the vest that incorporates sensor readings at multiple stages as well as the GSR upper back patch can be given a design patent. This prevents third parties from using a similar design for similar products.

When the devices are used, long term user data of individual patients gets collected. This can be made into a proprietary dataset that can be used anonymously to train future products as well as create an edge over upcoming competition.

3.1.2 Local vs. Global IP Process

In Sri Lanka, intellectual property is administered by the National Intellectual Property Office (NIPO) , but you can file a single international patent application from Sri Lanka, which acts as a placeholder in over 150 member countries. Sri Lanka is not a member of the Madrid Protocol. Therefore, you cannot file a single international trademark; you must file directly in the local IP office of each target country to legally protect and license your brand name globally.

3.1.3 FTO(Freedom To Operate)

This makes sure that the current design of a product does not collide with previous IP's held by other third parties. The clearance of FTO allows the device to be sold , marketed and distributed without external parties' IP's being violated. Therefore the obtaining of this is important for the commercial success of the product.

3.2 Regulatory Considerations

3.2.1 NMRA

The NMRA process takes at least 6 months and it entails registering through a local marketing authority holder, obtaining a sample import license and submitting a full product dossier.

3.2.2 FDA

The most important fact that we must be cautious about is if we are marketing this product as a clinical solution or a wellness product. The latter is easier as the FDA has a general wellness exemption clause which allows the product to be sold without extensive screening.

Claiming the device as a general wellness device prohibits the marketing of the device as a heatstroke diagnosis, prevention or mitigation device and instead as a "system that monitors and evaluates the risk of heatstroke occurrence in occupational and athletic settings".

If we were to claim diagnosis, prevention or mitigation of heatstroke, we have to comply with FDA class 2 medical device approval process formalities which includes implementing a full ISO13485 Quality Management System as well as going through Clinical trials, and waiting 6-18 months for FDA/NMRA approval . Therefore, going forward with a General Wellness product is the most practical solution.

3.3 Business Model Considerations

3.3.1 Concept 1: B2B Occupational Safety (Hardware-as-a-Service)

Targeting large-scale agricultural operations (e.g. commercial tea or rubber estates) and construction firms.

- Revenue: Lease the vests at a low upfront cost and charge a monthly subscription (Software-as-a-Service) per worker for access to a centralized fleet-monitoring dashboard that alerts managers to thermal strain.
- Advantage: High customer lifetime value, recurring revenue, and companies will pay to avoid liability and optimize labor efficiency during extreme weather.

3.3.2 Concept 2: B2B Elite Athletics

Selling premium vests to professional sports teams (e.g. cricket, rugby) and high-performance training institutes.

- Revenue: High upfront hardware cost with a premium analytics software subscription.
- Advantage: Easier to prototype for a small, highly controlled user base. High margins, but a very small total addressable market.

3.3.3 Concept 3: B2C Consumer Wellness

Selling directly to individual consumers, runners, and hikers via e-commerce.

- Revenue: One-off hardware sale, potentially with a basic free app.
- Advantage: Massive market size.
- Disadvantage: Requires a lot of marketing spending to acquire customers, and consumer hardware has very low profit margins.

3.4 Screening Summary

Table 2: Concept Screening Matrix (Pugh Method). Concept A = B2B Occupational Safety (HaaS), Concept B = B2B Elite Athletics, Concept C = B2C Consumer Wellness, '+' = the concept is better than the reference on that criterion, 'S' = the concept is the same as the reference on that criterion, '-' = the concept is worse than the reference on that criterion

Screening Criteria	Reference	Concept A	Concept B	Concept C
IP Freedom-to-Operate	S	S	S	S
Regulatory Feasibility	S	+	S	S
Business Model Viability	S	+	−	−
Technical Feasibility	S	S	+	−
Sum of Positives	–	2	1	0
Sum of Negatives	–	0	1	2
Sum of Sames	–	2	2	2
Overall Rating (Net)	–	+2	0	−2
Continue? (Go/No-Go/Revisit)	–	**Go**	Revisit	No-Go

Concept A (B2B Occupational Safety) scored highest on the screening matrix. It uses a Hardware-as-a-Service subscription model, which creates steady, recurring revenue.

It has a major regulatory advantage because it qualifies for the FDA general wellness exemption. Selling directly to employers avoids complex clinical testing and medical claims.

In addition, large agricultural and construction companies are eager to invest. They want this system to lower liability risks, protect workers, and improve job site efficiency.

Concept B (B2B Elite Athletics): Rated as a Revisit. It has good technical feasibility with a small, controlled user base for prototyping, but the market is too small to be a primary commercial path. It may be revisited later as a premium market segment.

Concept C (B2C Consumer Wellness): Screened out as a No-Go. It struggles with weak business model viability due to high customer acquisition costs and low hardware margins, plus technical challenges in achieving mass-manufacturing reliability at consumer prices.

4 Concept Testing

4.1 Prototyping

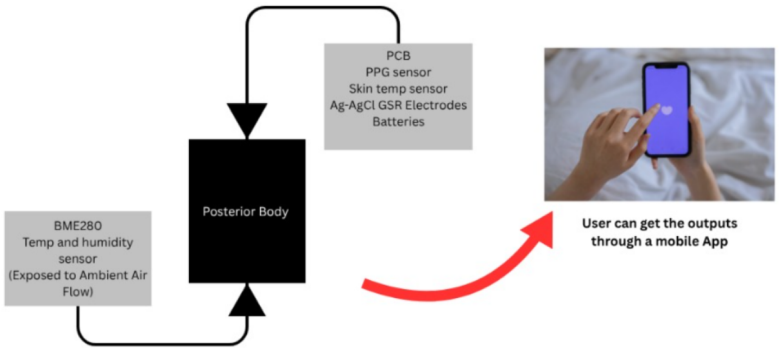


Figure 2: Concept map of the expected prototype

4.2 Testing Plan

To validate the multi sensor heatstroke monitoring system engineered to prevent heat, related illness during high physical exertion by tracking skin temperature, sweat dynamics, ambient heat index and severe tachycardia, a structured prototyping and testing framework was executed in accordance with the Biodesign process.

4.3 Bill of Materials Component Technical Advantages

The architecture combines physiological and environmental sensing to eliminate false readings caused by isolated sensor limitations.

Component Class	Selected Hardware	Technical Purpose	System Advantages & Justification
Core Microcontroller	ESP32-WROOM -32	Central data acquisition, DSP filtering, and wireless telemetry.	Dual core processing allows simultaneous high frequency I2C sensor polling and real time Bluetooth LE / Wi-Fi streaming without dropping data packets.
Heart Rate / Tachycardia Sensor	Integrated Optical PPG (Green Light LED) MAX30105	Continuous heart rate monitoring to detect exertion-induced tachycardia.	Green light spectrum offers superior signal to noise ratio and movement resistance over red/IR wavelengths during high impact physical activity.
Sweat & GSR Sensor	Ag-AgCl Electrodes	Measures Galvanic Skin Response to track changes in skin electrical conductivity.	Highly sensitive to rapid sweat gland activation and subsequent anhidrosis prior to severe heatstroke.
Skin Temperature Sensor	Medical-Grade IR / Contact Sensor (MAX30208 / MLX90614)	Monitors localized skin heating approaching critical 40 C thresholds.	High accuracy (+/- 0.1 C)provides precise tracking of core heating without delay from low cost thermistor drift.
Environmental Sensor	BME280 / DHT31	Measures ambient temperature and relative humidity.	Computes real time local Environmental Heat Index to cross reference physiological stress against ambient climate.
Enclosure & Optical Seal	3D-Printed Housing with Matte Black Foam Gasket	Structural housing at the upper thoracic / C7 anatomical region.	Ambient Light Shielding: Extended housing edge and matte black rubber gasket block external light leakage into the PPG photodiode.

Figure 3: Bill of Materials Component Technical Advantages

4.4 Prototyping Strategy

4.4.1 Stage 1 (Multi Sensor Data Fusion)

- Hardware Architecture:

  ESP32 interfaced via I2C/SPI with the Green-LED PPG sensor, MF52AT NTC Thermistor, SHT31 humidity and temperature sensor and Ag-AgCl GSR analog conditioning circuit.

- Data Cross-Referencing Software:

  C++ firmware executing multi marker decision logic to cross reference GSR conductivity spikes against ambient heat index and rising skin temperature, preventing salt crystal false positives.

- Power and Local Feedback:

  Integrated 3.7V LiPo battery with TP4056 charging IC and onboard piezo buzzer for immediate threshold alerts.

4.4.2 Stage 2 (Ergonomic Patch and Vest Housing)

- Anatomical Placement:

  Upper thoracic / C7 cervical spine patch placement, offering optimal contact stability, low motion disruption, and high wearability during exercise.

- Enclosure Design:

  Ultra-thin, ventilated CAD enclosure rendered in SOLIDWORKS and 3D-printed using flexible TPU / lightweight PLA.

- Skin Interface:

  Matte-black rubber gasket ring maintaining continuous light pressure against skin to block ambient light interference and secure electrode contact.

4.5 Comprehensive Testing Plan

4.5.1 Benchtop Sensor Calibration Cross-Talk Verification (Controlled Environmental Testing)

- Objective:

  Validate individual sensor accuracy and verify multi-sensor software cross-referencing.

- Methodology:

  Place the SHT31 temperature and humiduty sensor inside a stabilized environmental chamber, modulating the temperature from 25°C to 45°C. Inject synthetic GSR signals to test software suppression of false-positive alarms caused by salt crystallization.

- Pass/Fail Criteria:

  Temperature accuracy within +/- 0.2°C, complete software filtering of isolated GSR anomalies when ambient heat index and heart rate remain normal.

4.5.2 Motion Artifact and Optical Shielding Stress Test (Active Exertion Testing)

- Objective:

  Verify green light PPG tracking stability and ambient light shielding during physical movement.

- Methodology:

  Subjects perform dynamic athletic movements (running, jumping) under direct outdoor sunlight while wearing the C7 patch. Compare heart rate metrics against a standard chest strap monitor.

- Pass/Fail Criteria:

  Optical PPG signal maintains heart rate accuracy within +/- 4 bpm during continuous movement; zero optical saturation from sunlight leakage.

4.5.3 In-Situ Field Validation and Emergency Alarm Delivery (Simulated Heat Exertion Protocol)

- Objective:

  Confirm full system endurance (¿ 4 hours) and real time wireless alarm delivery under high thermal stress.

- Methodology:

  Athletes wear the prototype during regulated heat-acclimation training sessions. Program simulated threshold breaches to verify local audio alarms and BLE push notifications.

- Pass/Fail Criteria:

  Local buzzer actuation and mobile push alert delivery within ¡ 2 seconds of threshold breach. The device maintains secure skin contact without causing skin irritation or motion discomfort.

4.6 Wearable System Architecture and Hardware Distribution

- To maintain a lightweight, ergonomic profile during high exertion, hardware is strategically distributed across the posterior body. The Primary Sensor and Processing Node at the posterior C7 spine region houses a custom Main PCB with an ESP32 MCU and rechargeable batteries. It integrates a green light PPG sensor for tachycardia detection, a direct contact skin temperature sensor tracking spikes toward 40°C, and Ag-AgCl GSR electrodes to monitor sweat gland failure (anhidrosis).

- The Environmental Sensing Node is isolated on the outer fabric boundary exposed to ambient airflow. By placing the SHT31 sensor away from trapped body heat, it accurately measures ambient temperature and relative humidity to compute the true local heat index.

- The Wireless Data and User Output Interface eliminates heavy onboard displays to keep the vest thin and flexible. Telemetry from the posterior hub Streams via Bluetooth LE or Wi-Fi to a mobile application, delivering real time metrics and immediate emergency heatstroke push alerts.

5 Final Concept Selection

5.1 Killer Risks

Before scoring, leading concepts were evaluated for fatal technical and physiological flaws:

- Motion Artifacts: PPG sensors suffer heavily from motion noise during physical activity.
  - Mitigation: Using green light (MAX 30105) instead of red/IR (MAX 30102) for the PPG to provide the sharpest signal-to-noise ratio during daily movements, considering an anatomical location which suffers less from motion artifacts.
- Ambient Light Interference: External light leaking into the PPG sensor corrupts heart rate data.
  - Mitigation: Extending the housing 0.5mm to 1mm further than the body.
- Software/Sensor Cross-talk: Relying solely on one physiological marker (like skin resistance),without considering other conditions' interference fails during extreme conditions.
  - Mitigation: Implementing a software fix that strictly cross-references GSR spikes with ambient humidity and core temperature rising toward 40°C.
- The Diagnostic Reclassification Trap:
  - Explicitly marketing the vest as a device that "detects," "prevents," or "diagnoses" heatstroke legally reclassifies it as a Class II medical device under both FDA (US) and NMRA (Sri Lanka) regulations. This forces the startup into 12 to 24 months of clinical trial validation, ISO 13485 certification, and extensive regulatory fees before generating any revenue.

5.2 Selection Matrix

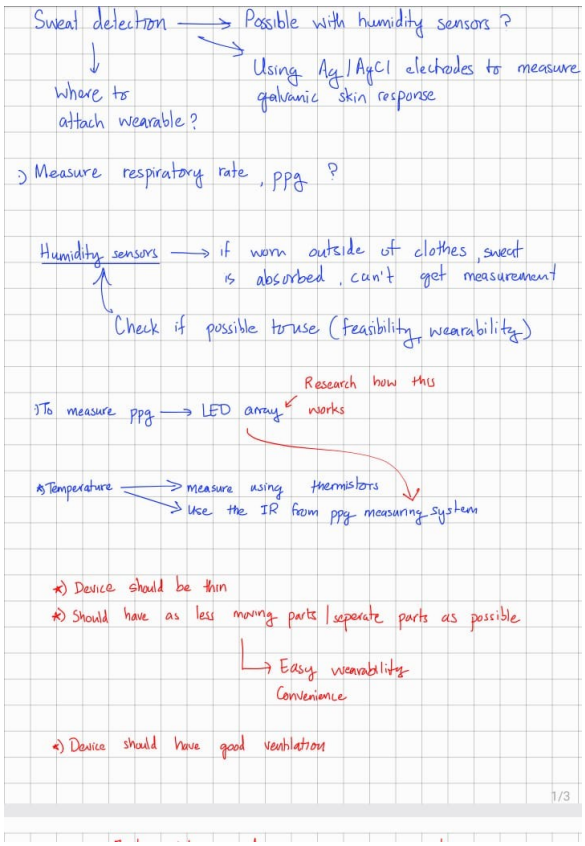
Requirement (Need Criteria)	Weight	Standalone GSR Strap (Baseline)	Smart Vest (With optimal sensor placements throughout the vest)	Multi-Sensor C7 Patch (GSR + IR + Green PPG)	Multi-Sensor Armpit Wearable (GSR + IR + Green PPG)
Provides continuous tracking during high motion	4	0	1	1	-1
Immune to false readings from salt crystals/flooding	5	0	1	0	-1
Measures multiple physiological markers (cross-referencing)	4	0	1	-1	1
High wearability (thin, few moving parts, vent, tilated)	3	0	1	-1	-1
Accurate ambient light blocking for continuous HR	3	0	1	1	1
Final Score		**0**	**19**	**6**	**-5**

Figure 4: Selection Matrix

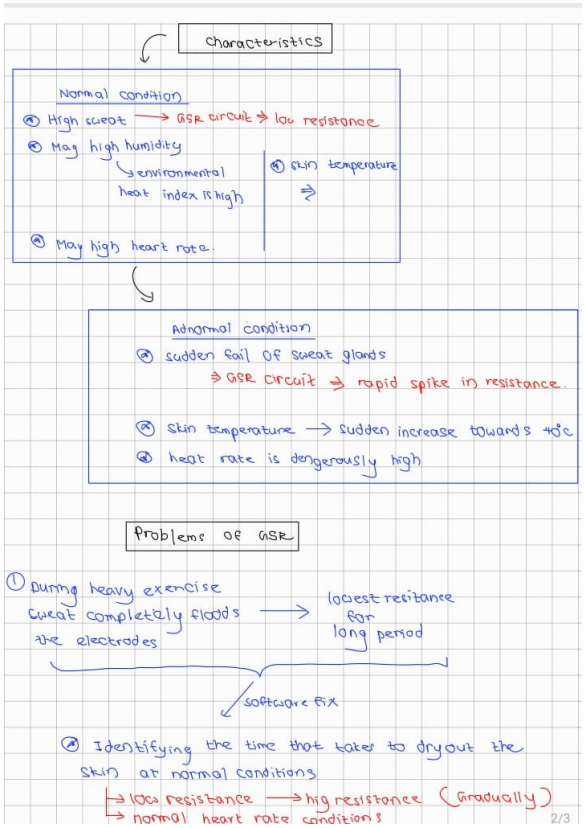
5.3 Final Concept Justification

Based on the highest score in the Pugh matrix, the Smart Vest with Multi Sensor arrangement is the final selected concept for development, as it successfully resolves the identified killer risks while meeting all criteria of the need statement.

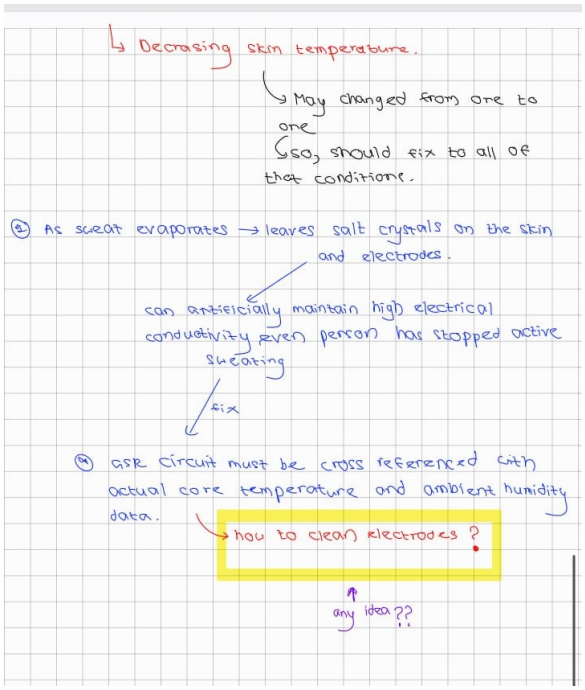
6 Brainstorming and Concept Map Evidence



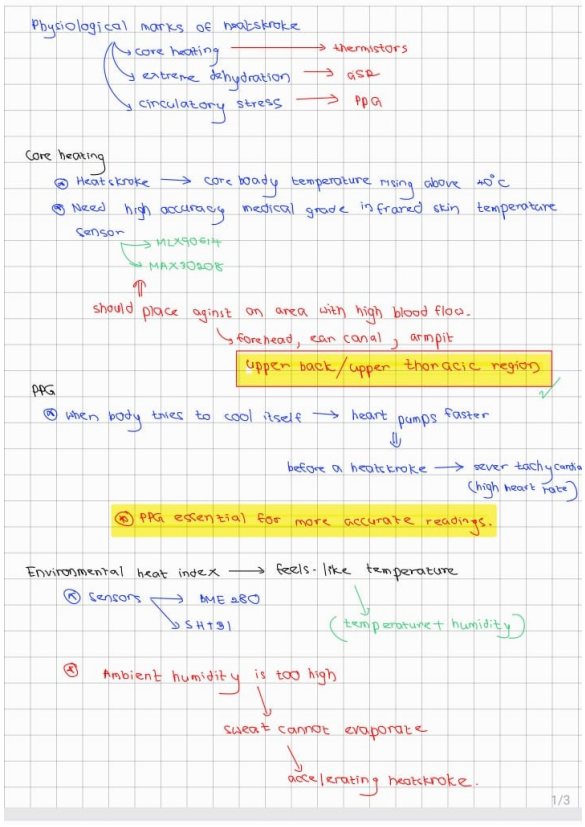
(a) Sensor Idea



(b) GSR Characteristics

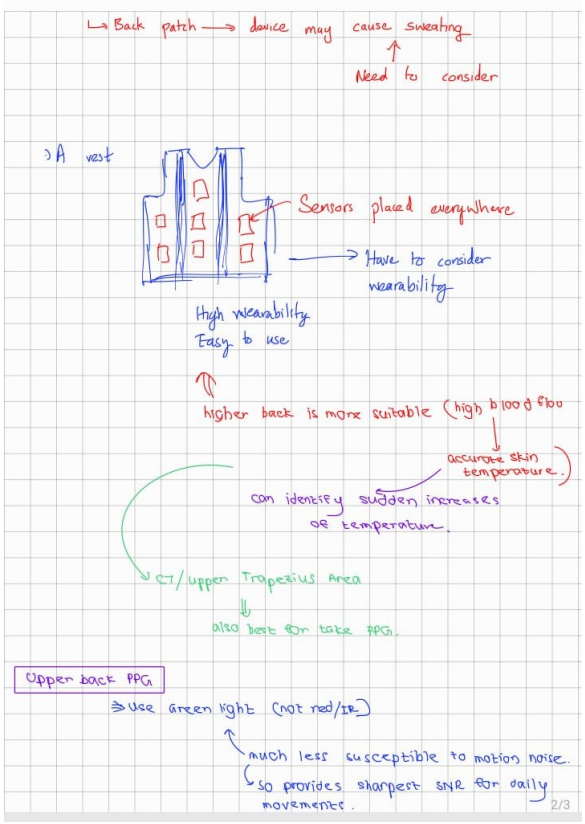


(c) GSR Limits

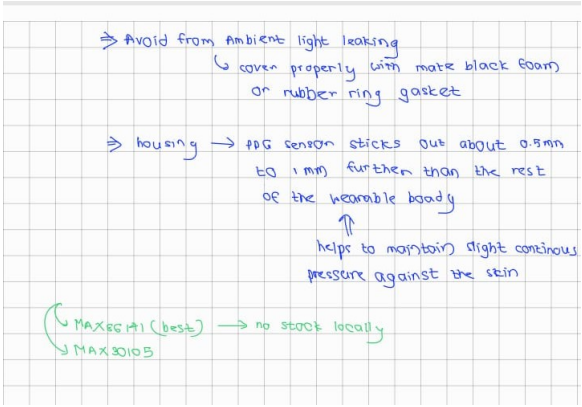


(d) Physiology

Figure 5: Ideation notes for physiological monitoring device, ordered sequentially (Part 1 of 2).



(e) Placement



(f) Housing

Figure 5: Ideation notes for physiological monitoring device, ordered sequentially (Part 2 of 2).

References

[1] Yock, P. G., Zenios, S., Makower, J., Brinton, T. J., Kumar, U. N., Watkins, F. T., and Denend, L. (2015). Biodesign: The process of innovating medical technologies (2nd ed.).